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Directed High Energy Radiation and Particle Beams Generated Using Extreme Magnetic Fields

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14. ABSTRACT The multi-PW, multi-beam laser facilities that are currently under construction will offer a unique capability of driving multiple current filaments in a dense plasma that can sustain extreme magnetic fields. The main goal of the project is to investigate how such a capability can be leveraged to create previously unattainable extreme conditions for delivering directed high-energy radiation and particle beams. We will carry out a numerical study of the dynamics of multiple megatesla magnetic field filaments driven by ultra-high intensity laser pulses in solid-density plasmas and the impact of the resulting configurations on quantum electrodynamics processes leading to generation of gamma rays and matter and antimatter. We will also conduct a feasibility study for direct experimental detection of the laser-driven MT magnetic fields using novel capabilities that will become available at the next-generation laser facilities, which would be critical for validating the regimes of interest experimentally. The ultimate goal of the project is to develop fundamental understanding of how multiple laser beams of extreme intensity interact with matter in order to develop experiments that would access the most promising regimes for gamma ray and pair production at the next generation laser systems, including ELI.					
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Final Technical Report

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PI Name:	Alexey Arefiev 512-589-8500, aarefiev@ucsd.edu

1 Accomplishments

This was a five-year project that started in 2017. The project was led by Prof. Arefiev at UC San Diego. The primary goal of the project was to identify promising mechanisms for generating directed high-energy radiation and particle beams with the help of ultra-high-intensity laser pulses. The project was primarily computational. High-intensity laser-plasma interactions and the resulting γ -ray emission are simulated using particle-in-cell (PIC) codes. These codes provide a kinetic description of the plasma. Such a description is computationally intensive (compared to a fluid description), but it is essential for correctly capturing the physics that involves a specific population of plasma electrons. In our project, we primarily used large PIC simulations performed on supercomputers.

Over the entire duration of the project, project participants published twenty four peer-reviewed articles [1–24], exploring various aspects of electron acceleration, ion acceleration, γ -ray generation, and production and acceleration of positrons. The focus of the computational research performed for this project was on those regimes that can lead to experiments in the near future at high-power high-intensity laser facilities. There are several interconnected research areas of the project that are worth highlighting in this regard: direct laser acceleration in dense plasmas; γ -ray generation; and positron generation from light alone. The work in the areas have resulted in the following significant results:

- Development of a new regime for direct laser acceleration in dense plasmas that leads to generation of a dense ultra-relativistic electron beam with energies in the 100 MeV to 1 GeV energy range at currently available laser intensities;
- Development of an approach for efficient γ -ray generation from dense laser-irradiated plasmas, where several percent of the laser energy is converted into a beam of multi-MeV photons;
- Discovery and development of a new regime for positron generation from light alone and their subsequent acceleration in laser-irradiated plasmas, with the 10^9 total positron yield for the currently available laser intensities.

The concepts developed for electron acceleration in dense plasmas are already being tested experimentally [16]. There are also several experiments that are being planned for implementing the key ideas for efficient γ -ray generation. The research on positron creation and acceleration has also produced unexpected ideas that pave the way to near-term experimental implementation. Specifically, the PI has submitted a new AFOSR proposal that focuses specifically on high-efficiency directed positron generation through binary photon collisions in laser-driven plasmas.

The results of our research were disseminated by publishing them in peer-reviewed scientific journals and by presenting them at conferences and meetings. In total, we published twenty four peer-reviewed articles [1–24], communicating most of our findings to the research community. In order to reach a broader research community and even general public, we employed press releases and news articles. For example, the research findings published in Refs. [15] and [18] were highlighted by multiple news outlets in multiple languages. A comprehensive list of the news articles can be found on the journal websites for these two articles. The results were also presented at least one major conference each year. The default was the Annual Meeting of the American Physical Society Division of Plasma Physics. In 2021, our work was recognized with an invited talk that was given by Y. He.

2 Impacts

2.1 Development of the principal discipline(s) of the project

The explosive increase in the number of high-power, high-intensity lasers that are now commercially available is a testament to how well humanity has mastered optical photons, including their generation using optical materials. The same cannot be said about γ rays whose energy is a million times greater ($\sim$ MeV), but the wavelength is a million times shorter ($\sim$ 0.01 Å). An efficient and compact γ -ray source is yet to be experimentally demonstrated. There is however a lot of interest in developing such sources because multiple applications, such as active interrogation, imaging of dense high-Z materials, and non-destructive testing of materials, need gamma-rays.

The project has contributed to the development of a novel concept that makes it possible to efficiently convert the energy of a high-intensity laser into a directed beam of energetic γ rays using a dense plasma. The conversion efficiency is at least several percent, which is a dramatic improvement over other concepts.

The project has also contributed to the development of two new approaches for producing matter and antimatter from light alone using lasers of currently available or soon to be available intensities. Previous research suggested that currently unattainable intensities are a must, so our results are an important breakthrough that opens the door to near-term experimental campaigns. The developed concept makes it possible to generate positrons from light alone inside a laser-irradiated plasma and then accelerate them to produce a directed positron beam.

2.2 Other disciplines

The new concept for the γ -ray generation is potentially important for such applications as active interrogation, imaging of dense high-Z materials, and non-destructive testing of materials.

The new concept for positron generation and acceleration is suitable for producing ultra-short relativistic positron beams. Such beams are in demand for positron annihilation lifetime spectroscopy [25] – a technique for non-destructive detection of sub-micron defects during manufacturing of high-value components [26]. Therefore, our project can potentially lead to the development of a useful tool, in the form of an ultra-short relativistic positron beam, for applications.

2.3 Impact on the development of human resources

The project has provided invaluable training in the field of high-intensity laser-matter interactions to multiple UC San Diego postdocs (O. Jansen, D. Blackman), graduate (T. Wang, Y. He, I.-L. Yeh, K. Yelshibekov) and undergraduate (K. Chin, S. Kojima) students. T. Wang obtained his Ph.D. in Engineering Physics in 2020. Y. He is scheduled to graduate in 2023 and I.-L. Yeh is scheduled to graduate in 2024. A significant portion of T. Wang's dissertation is the research performed for this project [1, 5, 7, 11, 13, 15, 16, 19]. The entire dissertation of Y. He is based on the research performed for this project [18, 20, 22, 24]. Student training in the field of high-intensity laser-matter interactions helps prepare qualified workforce for domestic national laboratories. All the students trained through this project learned to run and post-process supercomputer simulations, which is a valuable skill outside of the direct area of research.

2.4 Impact on teaching and educational experiences

Some of the topics, particularly regarding the generation of matter from light alone, have been highlighted in an undergraduate class on nuclear energy taught by the PI at UCSD.

2.5 Impact on physical, institutional, and information resources that form infrastructure

Nothing to report.

2.6 Impact on society beyond science and technology

Nothing to report.

3 Changes

There were no changes to our approach, so there is nothing to report. Last year the project was in the “no cost extension” mode, as we tried to wrap up as many things as possible. We were successful in that regard.

3.1 Changes in approach

Nothing to report.

3.2 Problems or delays

Nothing to report.

3.3 Expenditure impacts

Nothing to report.

4 Technical updates

The following section provides a detailed technical update on two major areas of our research: direct laser acceleration and γ -ray generation using a single laser beam and creation of positron beams from light alone.

4.1 Direct laser acceleration and γ -ray generation using a single laser beam

The PI's group has investigated multiple aspects of direct-laser acceleration (DLA) and photon emission in a laser-driven-plasma magnetic field [1, 4, 5, 7, 13, 15, 16]. What follows is a review of our recent accomplishments and relevant findings.

Generation of a strong quasi-static magnetic field via relativistic transparency: A plasma can generate and sustain a very strong magnetic field, but this strong field requires a high electron-current-density j in the plasma. Due to the fact that the electron velocity cannot exceed the speed of light c , there exists a hard upper limit, $|j| < |e|n_e c$, set by the electron density n_e in the target. The target has to be transparent to the irradiating laser pulse in order for the pulse to drive the volumetric plasma current and the resulting magnetic field. This requirement sets an upper limit for the electron density. If the plasma electrons are nonrelativistic, then the corresponding cutoff density that is often referred to as the critical density is only a function of the laser wavelength λ_0 : $n_{cr}[\text{cm}^{-3}] \approx 10^{21}/(\lambda_0[\mu\text{m}])^2$. At high laser intensity, the irradiated electrons become relativistic, which relaxes the constraint on n_e . The cutoff increases with laser amplitude, so the plasma remains transparent for $n_e < a_0 n_{cr}$, where

$$a_0 \approx 0.85 I_0 [10^{18} \text{W}/\text{cm}^2]^{1/2} \lambda_0 [\mu\text{m}] \quad (1)$$

is the normalized laser amplitude. This effect can be understood as a consequence of the effective increase of the electron mass. This regime requires $a_0 \geq 1$. To quantify the current density j , we use a dimensionless parameter [13]

$$\alpha \equiv \pi \lambda_0^2 |e| j / m_e c^3. \quad (2)$$

The current of relativistic electrons in a dense plasma with $n_e > n_{cr}$ corresponds to $\alpha > 1$, since $\alpha \approx 10$ for $j = |e| n_e c$ and $n_e = n_{cr}$.

The described increase of the cutoff density is a manifestation of a phenomenon called relativistically induced transparency [27–29]. This phenomenon of relativistic transparency is central to the research performed for the project, as it allows a high-intensity laser to propagate through a dense plasma and drive a strong electron current. In the example in Fig. 1, a high-intensity laser beam ($a_0 \approx 190$) easily propagates through a plasma-filled channel with $n_e = 20n_{cr} \gg n_{cr}$. The laser pulse (not shown to improve visibility) drives a strong longitudinal current shown in blue in Fig. 1(a). The resulting MA current that is distributed over the cross-section of the laser beam [$0.1 \text{ MA}/\mu\text{m}^2$ corresponds to $\alpha \approx 18$] generates and sustains a volumetric slowly-evolving azimuthal magnetic field whose maximum strength is 3 GG. This amplitude is only one order of magnitude lower than the amplitude of the oscillating laser magnetic field B_{laser} .

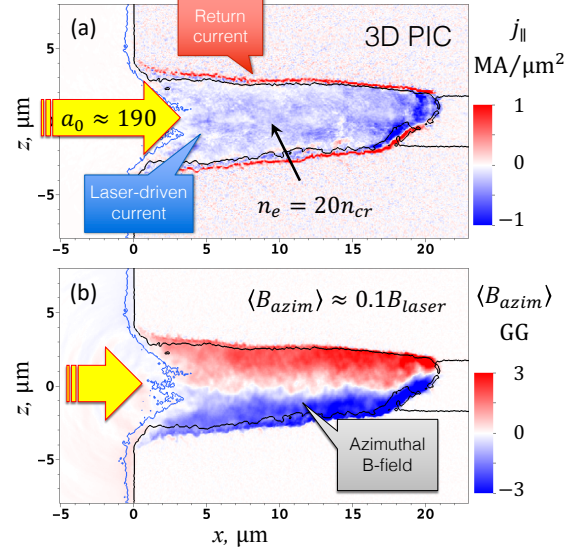


Figure 1: A laser with $a_0 \approx 190$ propagates through a plasma-filled channel, $n_e = 20n_{cr}$, (a) driving a strong current and (b) generating a static magnetic field. The angular brackets indicate time-averaging.

Direct laser acceleration assisted by the plasma magnetic field: A laser with $a_0 \gg 1$ can generate forward-directed, ultra-relativistic electrons. The energy is transferred from the transverse laser electric field $\mathbf{E}_{\perp}$ to transverse electron oscillations and then it is redirected by the laser magnetic field towards the forward-directed motion. The recipe for generating energetic electrons is to create the conditions where the transverse electron velocity $\mathbf{v}_{\perp}$ remains antiparallel to $\mathbf{E}_{\perp}$ over an extended period of time despite the oscillations of $\mathbf{E}_{\perp}$ (a review of direct laser acceleration is given in Ref. [30]).

We found that the plasma magnetic field driven by the laser can create such conditions [13, 31]. The magnetic field radially confines the electrons that are accelerated and pushed forward by the laser after being injected into the laser beam from ambient plasma (see Fig. 2). The radial deflections change the orientation of the transverse electron velocity $\mathbf{v}_{\perp}$ with respect to $\mathbf{E}_{\perp}$. If the corresponding frequency (betatron frequency) is comparable to the Doppler-shifted frequency of the laser, then $\mathbf{v}_{\perp}$ remains antiparallel to $\mathbf{E}_{\perp}$ even as the laser field and the electron oscillate. This mechanism of DLA assisted by the plasma magnetic field is shown in Fig. 2. The conditions are created by a laser with $a_0 \approx 80$ that drives a longitudinal electron current exceeding the non-relativistic Alfvén current [32], $J_A = m_e c^3 / |e| \approx 17 \text{ kA}$, by two orders of magnitude. This current, whose density corresponds to $\alpha \approx 17$ defined by Eq. (2), sustains a strong azimuthal magnetic field that confines the electrons and assists their energy gain from the laser. After travelling $50 \mu\text{m}$ with the laser, the electrons achieve a peak energy of $\varepsilon_e \approx 700 \text{ MeV}$.

Using a test-electron model developed by the PI, we found that the energy enhancement is a

threshold process (as opposed to a narrow linear resonance) controlled by a dimensionless combination α/a_0 [31, 33]. The electrons typically acquire some transverse momentum $p_0 \lesssim a_0 m_e c$ upon injection into the laser beam from the ambient plasma by the laser field $\mathbf{E}_\perp$. We have determined that, for a wide range of p_0 , electron energy gain approximately requires $\alpha > \alpha_{DLA} \equiv a_0/10$ [31]. It is convenient to re-cast this condition in terms of n_e by assuming that $j \approx |e|n_e c$, which gives a lower limit on the electron density: $n_e/n_{cr} \gtrsim 10^{-2} a_0$. The direct laser acceleration assisted by the plasma magnetic field then requires a strong current that can only be maintained by a sufficiently dense plasma. The regime in Fig. 2 satisfies this condition with $n_e/n_{cr} = 2$ and $10^{-2} a_0 = 0.8$.

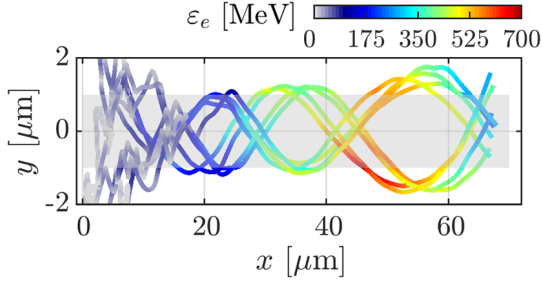


Figure 2: DLA of electrons assisted by a plasma magnetic field. The 3D PIC simulation is for a laser with $a_0 = 80$ irradiating a uniform plasma with $n_e = 2n_{cr}$ [13].

field. Static electric fields are present in the plasma, but they are typically much weaker than the magnetic fields at ultra-relativistic laser intensities [1].

Emission of γ rays: Electron acceleration associated with deflections in the magnetic field leads to emission of electromagnetic radiation. This process can be quantified using the expressions for synchrotron emission in a uniform magnetic field B by an electron with a relativistic factor γ [34]. The key parameter is the normalized acceleration $\eta = \gamma B/B_{crit}$, where $B_{crit} \equiv m_e^2 c^3 / \hbar |e| \approx 4.4 \times 10^{13}$ G is the magnetic equivalent of the Schwinger or critical electric field [35]. The emitted power scales as $P \propto \eta^2$. The spectrum emitted by an electron with energy ε_e has a peak at photon energy $\varepsilon_* \approx 0.4\eta\varepsilon_e$ (the parameter η is a simplified version of the quantum nonlinearity parameter χ_e [36] that involves electric and magnetic fields acting on the electron).

Similar physics is at play in XFELs, so we can make a meaningful comparison. EuXFEL uses 17.5 GeV electrons and a 10 kG magnetic field, so the corresponding value of the parameter that characterizes the photon emission is $\eta \approx 8 \times 10^{-6}$. The energy of the emitted photons is estimated as $\varepsilon_* \approx 50$ keV, which agrees within a factor of a few with the photon energies generated by EuXFEL.

The plasma magnetic field inside a dense laser-irradiated plasma can be at least five orders of magnitude higher. We find that $\eta \approx 0.013$ for a 1 GG field and 300 MeV electrons. Such a

The advantage of the discussed mechanism is that it is well-suited for generating relativistic electrons in a dense plasma that can sustain a strong magnetic field. The preferred density range is

$$10^{-2} \lesssim n_e/a_0 n_{cr} \ll 1. \quad (3)$$

The condition on the right is the condition for relativistically induced transparency. The condition on the left is the condition for electron energy gain. Our regime is the opposite of that used for the laser-wakefield acceleration since $1/\omega_{pe}$ is less than the laser pulse duration at high n_e . It is this aspect that enables the formation of the slowly-evolving magnetic

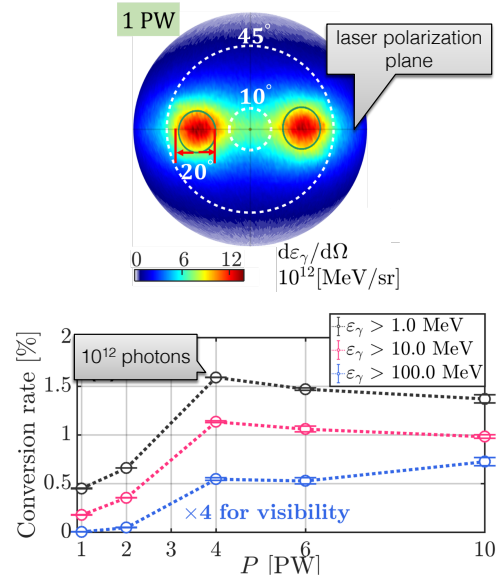


Figure 3: Photon emission ($\varepsilon_\gamma > 1$ MeV) projected onto a sphere and power scan for three different photon energy cutoffs [15]. The conversion rate is for a single lobe shown with a circle in the upper plot.

significant increase in η translates into a dramatic increase in the energies of emitted photons to $\varepsilon_* \approx 1.6$ MeV. This also means that the emission becomes a quantum process, as the photons can carry an appreciable fraction of the emitting electron's energy. Although the laser pulse also has a strong oscillating magnetic field, this field is unable to induce a similarly efficient photon emission process for forward moving electrons. The underlying reason is a well-known compensation of the magnetic field acceleration by the laser electric field (e.g. see Ref. [1]).

The described emission process is modeled using a Monte Carlo algorithm [37] that emits individual γ rays based on a corresponding synchrotron cross section. The photons are emitted along the momentum vector of the corresponding ultra-relativistic electron. The electron experiences the recoil following the photon emission, which reduces its momentum. The effect of the recoil is sometimes referred to as the force of radiation friction. The electron recoil must be taken into account for the regime of interest, because, as shown in recent publications by the PI's group, the force of radiation friction can have a nontrivial impact on electron dynamics during direct laser acceleration [4, 38]. The algorithm has been implemented into a particle-in-cell code EPOCH [39] that we use for our research. The algorithm has been benchmarked and extensively used for studies of high-intensity laser-plasma interactions. The reason a separate photon emission module is required is because the corresponding wavelength is six orders of magnitude shorter than the laser wavelength (the wavelength for $\varepsilon_\gamma = 1$ MeV is 10^{-3} nm), so it is not feasible to resolve it using the same field solver that propagates the laser pulse. The emission is treated as incoherent because the distance l between the electrons greatly exceeds the wavelength of the emitted radiation. For example, $l \approx 0.3$ nm for $n_e = 50n_{cr}$, where n_{cr} is calculated for an optical laser with wavelength $\lambda_0 = 1$ μ m. The wavelength of 1 MeV photons is two orders of magnitude smaller than l .

Our 3D PIC simulations have confirmed that a plasma with a strong laser-driven magnetic field can efficiently convert the energy of an optical laser into energetic γ rays. The resulting γ -ray beam has two well-collimated lobes separated along the laser polarization axis, shown in Fig. 3 for a 1 PW laser beam with $a_0 \approx 190$ irradiating a plasma-filled channel. The lower panel in Fig. 3 shows the energy conversion for three different groups of photons emitted into a single lobe. Even at 1 PW there are more than 10^{11} multi-MeV photons emitted into a single lobe, which accounts for 0.5% of the total energy in the laser beam. Both the photon number and the conversion rate are remarkably high and this is exactly what makes the considered approach so promising. For comparison, the energy conversion efficiency for laser-wakefield-based x-ray sources is roughly 10^{-5} (for 30 keV photons) [40]. This is due to the fact that the laser-driven wakefield acceleration requires a low density gas target, which limits the number of emitting electrons. In our mechanism, the photons are emitted over 100 fs, so the photon density is extremely high. As shown in Ref. [18], the density of photons with $\varepsilon_\gamma > 1$ keV in the emission region is $300n_{cr}$, exceeding the density of plasma electrons by two orders of magnitude. Some applications specifically require high photon densities, as in the case of electron-positron pair production from light alone [15, 18].

4.2 Creation of positron beams from light alone

Creation of electron-positron pairs via photon collisions, i.e. creation of matter from light alone, is often referred to as the Breit-Wheeler (BW) process [41]. The linear BW process involves two photons, with these photons typically being energetic γ rays. The nonlinear BW process involves multiple (more than two) photons, e.g. a GeV γ ray and multiple optical (eV-level) photons. These processes can be hard to observe in laboratory conditions. The nonlinear BW process has been observed once before by colliding a 46.6 GeV electron beam with a terawatt laser pulse [42]. The yield was less than one pair per shot. The linear BW process has only been observed via the use

of virtual photons, i.e. the photons quantized from the extremely strong electromagnetic fields generated by highly charged Au nuclei at ultrarelativistic speed [43]. There are two aspects that make the BW processes challenging to recreate: the smallness of the cross section and the high photon energy threshold. In the case of the linear BW process, the conservation of momentum and energy requires $\varepsilon_1 \varepsilon_2 > (m_e c^2)^2$, where $\varepsilon_{1,2}$ are the energies of the colliding photons, m_e is the electron mass, and c is the speed of light. This condition translates into a requirement to have a multi-MeV photon population.

High-power, high-intensity lasers have the potential to overcome these difficulties. Existing laser facilities are able to achieve laser intensities that exceed 10^{22} W/cm² [44–46], whereas the systems that are currently under construction are expected to approach intensities of 10^{23} W/cm² [45]. Most of the BW-related research has been focused on leveraging the anticipated ultra-high intensities that will exceed the 10^{23} W/cm² level to achieve a significant yield from the nonlinear BW process (e.g. see Refs. [47, 48]). The linear BW process has received less attention due to the perceived difficulties in creating adequate conditions [49]. The key difference between the nonlinear BW and the linear BW processes is that the nonlinear process is a collision of γ rays with laser photons, whereas the linear process is a collision of γ rays with other γ rays. The reason why the nonlinear-BW process requires a very high intensity is because it needs a high density of optical photons to achieve a measurable yield. In contrast to the nonlinear BW process, the linear BW process has no explicit laser intensity requirement. We have discovered two regimes that take advantage of this feature to achieve an appreciable yield via the linear BW process using currently available laser intensities.

One way to utilize high-intensity lasers for the linear BW process is by converting optical photons into γ rays. It has been shown that dense laser-irradiated plasmas can efficiently convert laser energy into γ rays. The laser first transfers its energy to plasma electrons, accelerating them to ultra-relativistic energies (hundreds of MeV). Energetic γ rays are then efficiently emitted by these relativistic electrons via synchrotron emission during electron deflections by strong laser and plasma fields [51–57]. As already discussed in this report, the PI has investigated together with his group multiple aspects of γ -ray emission in high-intensity laser-plasma interactions [7, 15, 16, 31, 53]. According to computational studies, the described mechanism can convert $\sim 10^{-2}$ of the laser energy into a beam of collimated multi-MeV γ rays at experimentally achievable intensity of 5×10^{22} W/cm² [15]. For comparison, the energy conversion efficiency for laser-wakefield-based x-ray sources (these use moderately relativistic laser intensities) is roughly 10^{-5} (for 30 keV photons) [40].

How to utilize the emitted γ rays for the pair-production is an open question. There has been increased interest in utilizing γ rays emitted during ultra-intense laser plasma interactions to develop setups that can be used to experimentally observe the linear BW process [15, 18, 50, 58–61]. The PI's group has actively worked on the topic of pair-production via the linear BW process during the AFOSR project covered in this report, publishing seven papers [1, 15, 17, 18, 20, 22, 24]. One general approach is to use two lasers, each irradiating its own target and generating a beam of γ rays [60]. The γ rays are then collided some distance away from the targets, as seen in Fig. 4. The appeal is that the pairs are created in vacuum. The drawback however is that, due to the divergence

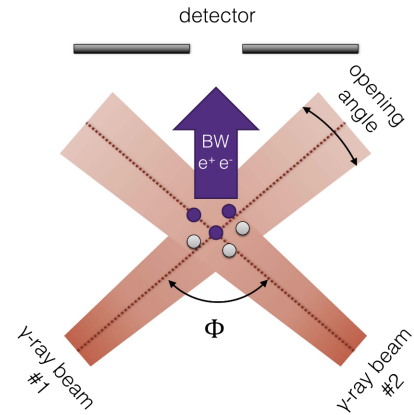


Figure 4: Schematic setup for pair production via the linear BW process by employing two laser-driven γ -ray beams [1, 50]. The beams collide some distance away from the laser-irradiated targets that produce them.

of the γ rays, the yield strongly depends on the distance from the targets to the collision location. In practice, this means that the distance should not exceed several hundred microns to achieve a yield that is more than a few pairs [15], which makes experimental implementation challenging. It is thus necessary to re-evaluate the concept rather than to pursue its further optimization.

Linear BW pair-production by two laser pulses inside a plasma: Prompted by the predictions of low yield, the PI's group at UCSD started looking for new approaches. In a major departure from previous research, we decided to ease the requirement that the pairs must be produced in vacuum, allowing them to collide inside a plasma. A possible downside of doing this is that processes other than the linear BW process can contribute to the positron yield, which can make the interpretation of measurements challenging. On the upside, this approach can mitigate the negative impact of the γ -ray divergence on the linear BW pair-yield.

A quantitative assessment of the linear BW pair yield inside a plasma is far from straightforward because the pairs are produced at the same time and location as the γ rays. High-intensity laser-plasma interactions and the resulting γ -ray emission are simulated using PIC codes. These codes provide a kinetic description of the plasma. Such a description is computationally intensive (compared to a fluid description), but it is essential for correctly capturing the physics. The γ rays are produced by a non-Maxwellian energetic tail of the electron distribution and the kinetic approach allows one to correctly compute the formation of the tail. In most PIC simulations, the photons are emitted by plasma electrons as individual particles, with the emission performed by a separate module [37]. In contrast to the nonlinear BW process that is already implemented in all major PIC codes, the linear BW process requires simulating binary photon collisions. As of today, none of the public releases

of major PIC codes are equipped to simulate the linear BW process in laser-plasma interactions. In order to make progress, we developed a post-processing module for computing the yield from the linear BW process. The module employs an algorithm that splits the PIC computational domain into smaller sub-domains. The photons emitted by plasma electrons within each sub-domain are represented by multiple collimated mono-energetic beams called beamlets. The algorithm computes the linear BW process yield by evaluating beamlet-beamlet collisions without the spatial integration over the interaction region, which makes calculations fast. This development was funded by AFOSR and the details of the algorithm were published in Computer Physics Communications [24].

The first setup that we investigated using the developed post-processing module is shown in Fig. 5. Like the setup in Fig. 4, it involves two lasers. However, it uses just a single target, so that the γ rays produced by each laser collide head-on inside this target without having an opportunity to diverge [18, 22]. As seen in Fig. 5(b), the photon density during the collision exceeds the density of electrons inside the target by an order of magnitude. Not surprisingly, the resulting yield from the linear BW process is remarkably high. Even at 2×10^{22} W/cm² in each beam, the photons produce

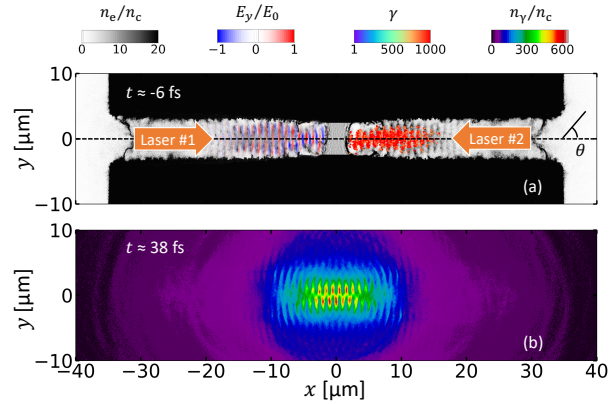


Figure 5: Generation of energetic photons in a structured target irradiated by two counter-propagating laser beams [18, 22]. (a) Electron density n_e (gray scale), transverse electric field of laser #1 E_y (color scale) and energetic electrons with $\gamma \geq 800$ accelerated by laser #2 (dots, colored by γ). The snapshot is taken at $t \approx -6$ fs prior to the laser collision, $t = 0$. (b) The density of photons with energy $\epsilon \geq 1$ keV, in units of the classical critical density n_c at $t \approx 38$ fs (after the laser collision).

over 10^8 pairs. There are two competing pair-production processes: the nonlinear BW process (collision of γ rays produced by one laser beam with the laser beam propagating in the opposite direction) and the Bethe-Heitler process (collision of γ rays with atomic nuclei). Our calculations show that the yield from these processes is lower by two orders of magnitude [18]. Therefore, the linear BW process in our setup is a dominant pair-production process, generating a significant yield at currently available laser intensities.

Even though this first result for pair-production inside a plasma has demonstrated a lot of potential of the new approach (Ref. [18] has already been cited 33 times per Google Scholar), the setup itself remains challenging for immediate implementation. It does require two laser pulses, so it is limited to a handful of laser facilities that have this capability. The use of a structured target seen in Fig. 5(a) might be necessary to keep the lasers aligned during propagation, with the pre-filled channel serving as an optical wave-guide for the laser beams entering it from both ends. Such targets have been fabricated at GA and deployed at a recent experiment at the Texas Petawatt [16]. The UCSD group was directly involved, providing computational and theoretical support.

Linear BW pair-production by a single laser pulse inside a plasma:

In our search for a simpler configuration, we discovered that a single laser pulse can also create conditions that are favorable for the linear BW process. In Ref. [20], we showed that a high-intensity laser pulse propagating through a dense plasma can facilitate emission of backward-directed γ rays in addition to the forward-directed γ rays that have been the primary focus of research in the laser-plasma community. An angularly resolved photon spectrum from a 2D PIC simulation seen in Fig. 6 is shown in Fig. 7, where the forward-moving photons are shown in Figs. 7(a)&(b) and the backward-moving photons are shown in Fig. 7(c). The high energy and number of the backward-directed photons offer an exciting possibility of producing pairs via photon-photon collisions using just a single laser.

The backward and forward photons are produced by two distinct electron populations. The fields that cause the electrons to emit are also different (the laser fields for the former and the plasma magnetic field for the later). As a consequence, the backward-directed photons are emitted ahead of the forward-directed photons, creating a self-organized photon collider. Figure 6 illustrates this for a 25 fs, 800 nm laser with a peak intensity of 3×10^{22} W/cm² propagating through a structured target. Figure 6(d) shows the density of photons with energy above 100 keV. This population consists of overlapping backward and forward moving photons [Figs. 6(h)&(g)]. The same physics is observed in simulations with a uniform target.

A calculation using our post-processing module [24] predicts a total yield of $N_{lin}^{BW} \approx 1.1 \times 10^7$

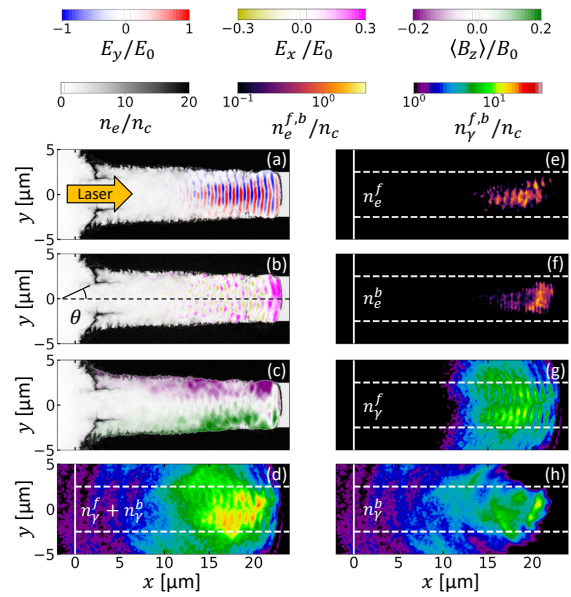


Figure 6: Generation of counter-propagating γ rays by a single laser. (a)&(b) Transverse, E_y , and longitudinal, E_x , electric fields normalized to $E_0 \approx 1.6 \times 10^{10}$ statV/cm. (c) Time-averaged magnetic field, $\langle B_z \rangle$, normalized to $B_0 \approx 16$ GG. (d) Density of emitted photons with $\varepsilon > 100$ keV. (e)&(f) Density of forward and backward moving electrons with $\chi_e > 0.03$ and $\gamma > 15$. (g)&(h) Density of forward and backward moving photons with $\varepsilon > 100$ keV.

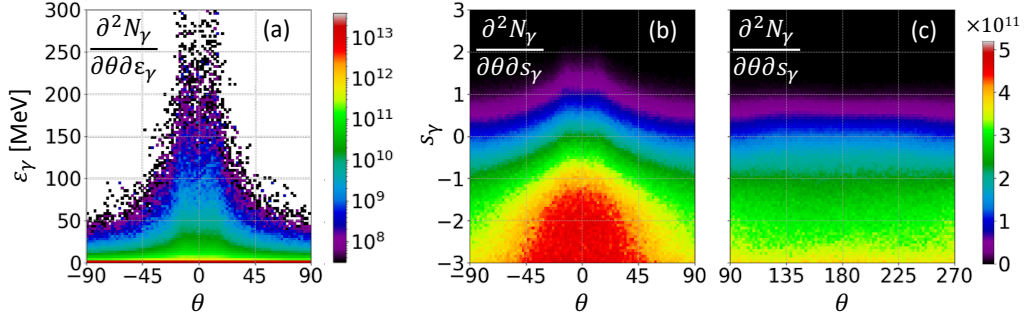


Figure 7: Energy-angle spectra of the photons emission from Fig. 6. (a)&(b) The distribution of the forward-directed photons on linear and logarithmic energy scales, where $s_\gamma \equiv \log_{10}(\varepsilon_\gamma [\text{MeV}])$. (c) The distribution of the backward-directed photons on a logarithmic energy scale.

electron-positron pairs. This number can be confirmed with a simple estimate. The cross-section for the linear BW process, $\sigma_{\gamma\gamma}$, peaks for a head-on collision of two 700 keV photons, with $\max(\sigma_{\gamma\gamma}) \approx 1.7 \times 10^{-29} \text{ m}^{-2}$. The density of photons with $\varepsilon > 700 \text{ keV}$ is $n_\gamma \approx 4n_c$, where $n_c \approx 1.7 \times 10^{27} \text{ m}^{-3}$ is the critical density for a laser with wavelength $\lambda_0 = 0.8 \text{ }\mu\text{m}$. Their total number is $N_\gamma \approx 4 \times 10^{13}$, where we assume that the size of the photon cloud in the third dimension is equal to the width of the laser beam $w_l \approx 5 \text{ }\mu\text{m}$ (our simulation is 2D). The backward-emitted photons are not collimated, so we assume that they leave the cloud after traveling a distance roughly equal to the radius of the laser beam, $w_l/2$. Putting these estimates together, we find that indeed the total yield is $N_{lin}^{BW} \approx \sigma_{\gamma\gamma} n_\gamma N_\gamma w_l/2 \approx 1.2 \times 10^7$.

The emission of colliding photons is closely connected to longitudinal electric and azimuthal magnetic fields that emerge in the plasma during laser propagation. Their structure is shown in Figs. 6(b)&(c). A relativistic electron experiencing deflection by electric and magnetic fields ($\mathbf{E}$ and $\mathbf{B}$) emits photons in the direction of its momentum. Therefore, the forward-directed photons are emitted by forward-moving electrons and the backward-directed photons are emitted by backward-moving electrons. The characteristic energy of these photons is $\bar{\varepsilon} \approx 0.44\gamma_e\chi_e m_e c^2$, where

$$\chi_e = \frac{\gamma_e}{B_{crit}} \sqrt{\left(\mathbf{E} + \frac{1}{c}[\mathbf{v} \times \mathbf{B}]\right)^2 - \frac{1}{c^2}(\mathbf{E} \cdot \mathbf{v})^2} \quad (4)$$

is a dimensionless parameter that quantifies electron acceleration. Here $\mathbf{v}$ is the electron velocity, $\gamma_e = (1 - v^2/c^2)^{-1/2}$ is the relativistic factor, and $B_{crit} \approx 4.4 \times 10^{13} \text{ G}$ is the magnetic field equivalent of the Schwinger (or critical) electric field [35]. Multi-MeV photon emission by a $\sim 100 \text{ MeV}$ electron requires $\chi_e \approx 0.02$, which implies very strong acceleration. In contrast, χ_e is three orders of magnitude lower in XFEL's ($\chi_e \approx 10^{-5}$ for 17.5 GeV electrons deflected by a 10 kG field). In our setup, the forward and backward moving electrons achieve the required value of χ_e in two different ways because χ_e strongly depends on the field orientation relative to the electron velocity $\mathbf{v}$.

Forward moving electrons are accelerated by the laser pulse itself and they emit because of deflections by the azimuthal plasma magnetic field that emerges in the plasma during laser propagation. Laser acceleration produces energetic forward moving electrons due to a forward push from the laser magnetic field. As a result of the forward motion, the contributions from the transverse laser $\mathbf{E}$ and $\mathbf{B}$ fields to χ_e nearly cancel each other out. The plasma $\mathbf{B}$ -field has no electric counterpart, so it is able to induce strong acceleration and it is the strength of this field that determines χ_e . The plasma field has to be in the GG range to reach the desired value of χ_e . This strength is

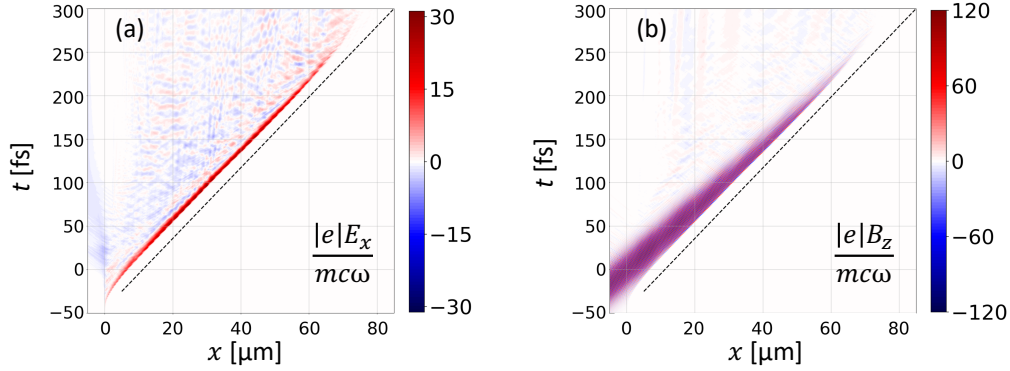


Figure 8: Time evolution of E_x (mainly plasma field) and B_z (mainly laser field) in the near-axis region in the PIC simulation shown in Fig. 6. A horizontal slice gives a field profile for a given time t . The fields are spatially averaged in the transverse direction over a region with $|y| < 1.5 \mu\text{m}$. The dashed black lines (the same in both panels) represent longitudinal motion with velocity $v_x = 0.82c$.

achieved by using a dense plasma that can sustain a sufficiently strong longitudinal electron current driven by the laser. In our simulation, the plasma electron density is higher than the classical cutoff density n_c . The laser is able to propagate through the plasma due to the relativistically induced transparency [28, 29] – the phenomenon where electron heating to relativistic energies raises the cutoff density and makes an opaque plasma transparent. Because of the high electron density, the quasi-static azimuthal plasma magnetic field in Fig. 6(c) is 3 GG or 20% of the oscillating laser magnetic field B_0 (in the absence of the target).

Backward moving electrons are generated by the longitudinal plasma electric field at the leading edge of the laser pulse and they emit because of their head-on collision with the laser. The longitudinal field E_x is formed by charge separation as the leading edge of the laser pulse sweeps up the electrons it encounters in its path while the ions are left behind. After initial forward acceleration, the swept up electrons slow down under the influence of the longitudinal field and then re-accelerate in the backward direction. As seen in Fig. 8, the E_x moves forward with the leading edge of the laser pulse and the population of the swept up electrons is constantly replenished. In our simulation, the amplitude of E_x is 25% of the laser amplitude, so this field is able to accelerate backward moving electrons to relativistic energies, with $\gamma_e \sim 100$. In contrast to the case of the forward-moving electrons, χ_e for the backward-moving electrons is determined by the fields of the laser. The reason for this difference is the lack of compensation between $\mathbf{E}$ and $\mathbf{B}$ of the laser. Even though the backward-moving electrons are less energetic than the forward-moving electrons, they are able to reach the value of χ_e needed for the emission of multi-MeV photons because they are deflected by a much stronger field (the laser fields are much stronger than the plasma magnetic field).

The calculated $\sim 10^7$ positron yield at an experimentally achievable intensity demonstrates feasibility of the single-laser setup. The setup is simple, as it only requires a single laser pulse with a reasonable peak intensity, so it can and should be considered for near-term implementation. We plan to focus on this during our next project.

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